INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

PACKING, MATERIAL AND PACKING PREFORMED (STUFFING TUBE FOR ELECTRIC CABLES)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

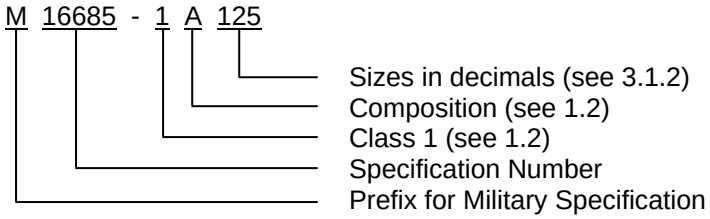
1.1 Scope. This specification covers rounded packing on spools or reels and preformed packing for electric cable seals in stuffing tubes.

1.2 Classification. Packings are of the following classes, compositions, and types, as specified (see table I, figures 1 and 2 and 6.2):

Class 1 - Rounded packing on spools or reels.
  Composition A - Retainer packing.
  Composition B - Sealing packing.
Class 2 - Preformed. (Made up from composition A and B packing (see 3.2).
  Type I - For deck and bulkhead stuffing tubes (see Table III).
  Type IA - For deck and bulkhead stuffing tubes (see Table III).
  Type II - For equipment stuffing tubes (see Table III).

1.3 Part or Identifying Number (PIN). The PINs to be used for packing acquired to this specification are created as follows:

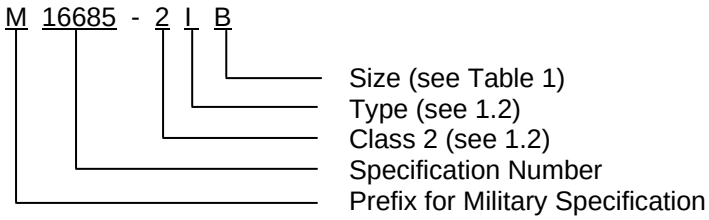
Class 1



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Class 2



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 Sampling Procedures and tables for Inspection by Attributes.

(Copies of this document are available from www.asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Class 1, rounded.

3.1.1 Material.

3.1.1.1 Composition A. The fiber shall be composed of polytetrafluoroethylene (expanded PTFE) graphite, and a high temperature lubricant. This product shall be asbestos free.

3.1.1.2 Composition B. Packing shall be composed of an admixture of cork particles, graphite, lubricant and a plastic binder enclosed in a woven cotton lattice jacket.

3.1.2 Sizes. Packing shall be furnished in rounded cross section in sizes 1/8 (.125), 3/16 (.188), 1/4 (.250), 5/16 (.313), 3/8 (.375), 7/16 (.438), and1/2 (.500) as specified (see 6.2).

3.2 Class 2, preformed.

3.2.1 Material.

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3.2.1.1 Piece1. Piece 1 shall be made of packing conforming to class 1, composition A, as shown on figures 1, 2, and 3.

3.2.1.2 Piece 2. Piece 2 shall be made of packing conforming to class 1, composition B, which may be made in one continuous piece in the form of a tight helix, or may be cut into individual rings (see figures 1, 2, and 4). If made of individual rings, the cuts shall be displaced approximately 90 degrees from the cuts in the adjoining rings. The cut in each top and bottom retainer rings shall be similarly displaced from the cut in the adjoining sealing component.

3.2.2 Coating. After molding, packing shall receive a thin coat of a suitable quick drying resin preparation. The coating shall be of such a nature that it will in no way affect the sealing properties of the class 1, composition B packing, when tested as specified in 4.4.

3.2.3 Sizes. Packing shall be of the sizes shown in table I and as specified (see 6.2).

Table I - Sizes.

Size	1/ d Inch	Dimensions				
		outside diameter Inches	inside diameter Inches	h-I (type I) Inches	h-IA (type IA) Inches	h-II (type II) Inches
A	3/16 (.188)	.750	.375	1.375 ± .063	1.375 ± .063	1.125 ± .063
B	3/16 (.188)	.875	.480			
C	3/16 (.188)	1.000	.610			
D	3/16 (.188)	1.125	.719			
E	3/16 (.188)	1.125	.778			
F	3/16 (.188)	1.187	.812			
G	1/4 (.250)	1.375	.911			
J	1/4 (.250)	1.500	1.037			
K	1/4 (.250)	1.625	1.120	2.625 ± .094	1.688 ± .063	1.375 ± .063
L	1/4 (.250)	1.687	1.232			
M	1/4 (.250)	1.875	1.375			
N	1/4 (.250)	1.937	1.474			
P	1/4 (.250)	2.062	1.586			
R	5/16 (.313)	2.187	1.681			
S	3/8 (.375)	2.562	1.820	4.625 ± .125	2.563 ± .094	2.125 ± .094
T	3/8 (.375)	2.750	2.030			
V	3/8 (.375)	2.875	2.142			
W	3/8 (.375)	3.000	2.280			
X	3/8 (.375)	3.125	2.427			
Y	3/8 (.375)	3.250	2.545			
Z	3/8 (.375)	3.437	2.673			

1/ Minimum diameter of size of piece 1 before molding.

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3.3 Resistance to accelerated aging. Packing shall withstand the test specified in 4.4.1 without hardening or undergoing any change which indicates deterioration.

3.4 Molding. The rounded, class 1, packing shall be capable of being readily molded or prefabricated into packing sets. This capability shall be determined by the test described in 4.4.3.

3.5 Performance. Packing shall be capable of remaining tight indefinitely, and shall have no deleterious effects upon the sheathing of electric cables or upon the metallic armor of electric cables, where used against water, air, or gas at pressures from atmospheric to 25 p.s.i.g. (.1724 MPa) at temperatures up to 212 °F (100 °C). (see 4.4.2).

3.6 Compressibility. Packing shall be formed under the minimum pressures specified in table II with the packing and molds maintained at a minimum temperature of 100 °F (37.8 °C). When tested as specified in 4.4.3, the height shall be not less than 80 percent of the original height. For packing, class 1, furnished in bulk, spools or reels, check compressibility of all diameter of cable and packings in mold sizes equal to stuffing tube sizes C, L, R, and V, respectively, in the above manner.

Table II - Molding pressures.

Type	Size	Minimum pressure (p.s.i. of annular area of packing)
I	A – J	600
I	K – Z	400
IA, II	A – R	600
IA, II	S - Z	400

4. VERIFICATION

4.1 Sampling for conformance inspection.

4.1 Sampling lot. All packing of the same class, composition, type, or size shall be considered a lot for purposes of acceptance inspection.

4.1.2 Sampling for examination. Sample sets or spools shall be selected at random in accordance with ASQ Z1.4 at Inspection Level I for the examination of 4.2.1. The Acceptable Quality Level shall be 2.5 percent defective.

4.1.3 Sampling for tests. Sample sets or spools shall be selected at random in accordance with ASQ Z1.4 at Inspection Level S2 for tests specified in 4.2.2.

4.1. 4 Sampling for compression test. A random sample of sets or spools shall be selected from each lot in accordance with 4.1.3 and subjected to the test specified in 4.3.3.

4.1.5 Sampling for production check test. Three yards of rounded packing and two sets of preformed packing from the first lot of each type and class and 54 in. (137 cm) of rounded packing and one set of preformed packing from every tenth lot shall be selected from the same class and type for the tests specified in 4.3. The samples shall be chosen so that over a period of time a representative selection of sizes will be sampled.

4.2 Conformance inspection.

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4.2.1 Visual and dimensional examination. Each of the sample sets or spools (see 4.1.2), shall be visually and dimensionally examined to verify compliance with this specification. Any set or spool in the sample containing one or more visual or dimensional defects shall not be offered for delivery and if the number of defective sets in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.2.2 Conformance tests. Unless otherwise specified conformance tests shall include the tests specified in 4.3 for each size, type and class of packing.

4.2.2.1 Compression test. Each of the sample sets or spools selected in accordance with 4.1.4 shall be tested in accordance with 4.3.3 to verify compliance with this specification. If any of the packing does not meet the requirements of this test, this shall be cause for rejection of the lot represented by the sample.

4.2.2.2 Production check test. The sample selected as specified in 4.1.5 shall be subjected to the tests specified in 4.3.

4.2.2.2.1 Action in case of failure. Lot acceptance and rejection will normally be on the basis of sampling, examination and tests specified in 4.1, 4.2.1, and 4.2.2.1 only, and acceptance will not be withheld pending receipt of the test report on the production check test samples. However, upon receipt of an unsatisfactory report on any sample chosen for production check tests, additional samples shall be selected from every subsequent lot offered for delivery. These samples shall be subjected to the tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. This additional testing shall be discontinued and lot acceptance returned to the normal basis when two successive lots have been accepted.

4.3 Test procedures.

4.3.1 Accelerated aging test. Specimens shall be suspended in an electrically heated oven and shall be subjected to dry circulating air at 200 °F (93 °C). for 1000 hours without hardening or softening or without changing in any other respect which will indicate that the material is deteriorating.

4.3.2 Performance test. This test shall be conducted in representative sizes of bulkhead stuffing tubes. Both plain and armored cable (including sheathing material of polychloroprene, impervious resins and thermoplastic resins) shall be used during the test. After the packing has been installed in the tubes, the gland units shall be drawn up snugly with a wrench 20 in. (50.8 cm) in length. An air pressure of 25 p.s.i.g. (.1723 MPa) for bulkhead stuffing tubes shall then be applied within the apparatus and any leakage that occurs from the tubes shall be noted and recorded. The air, with pressure still applied within the apparatus, shall be heated to a temperature of 212 °F (100 °C) by means of a water bath surrounding the compartments. These conditions shall be maintained for 1.5 hours and any leak which occurs from the tubes shall be noted and recorded. At the end of this time the apparatus shall be allowed to cool and, when cold, the leakage from the tubes shall again be recorded. The above conditions shall be repeated at least three times. On conclusion of the test, the packing shall be removed from the tubes and the cables examined for crimpage, deformation, or evidence of gross damage by chemical attack of armor or cable sheath. If any of these mentioned defects are in evidence, this shall be cause for rejection of the lot represented by the sample.

4.3.3 Compression test. The adequacy of the molding pressures, as specified in 3.6 shall be determined as follows: Measure the height dimension (h-I, h-IA or h-II of figures 1 and 2, as applicable) of a sample of preformed packing. Place the sample in the corresponding size mold and subject it to the pressure specified in 3.6 for a period of 3 minutes. Release pressure and measure the same dimension as soon as the set is removed from the mold.

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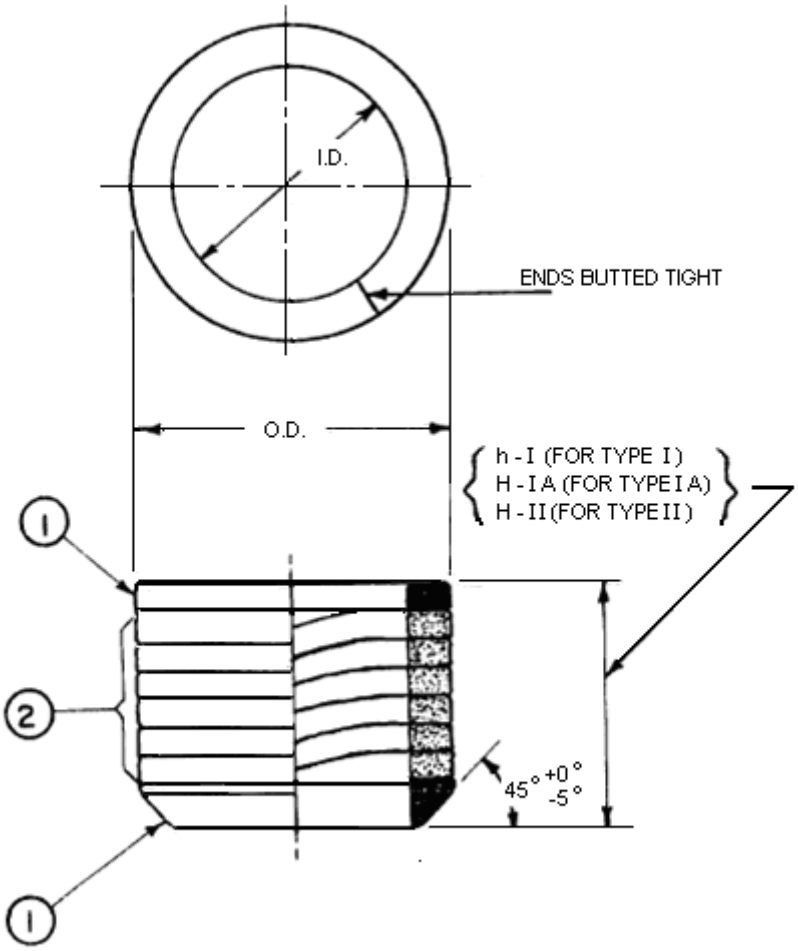


Figure 1 - Type I sizes (A to J)
Type IA (all sizes)
Type II (all sizes

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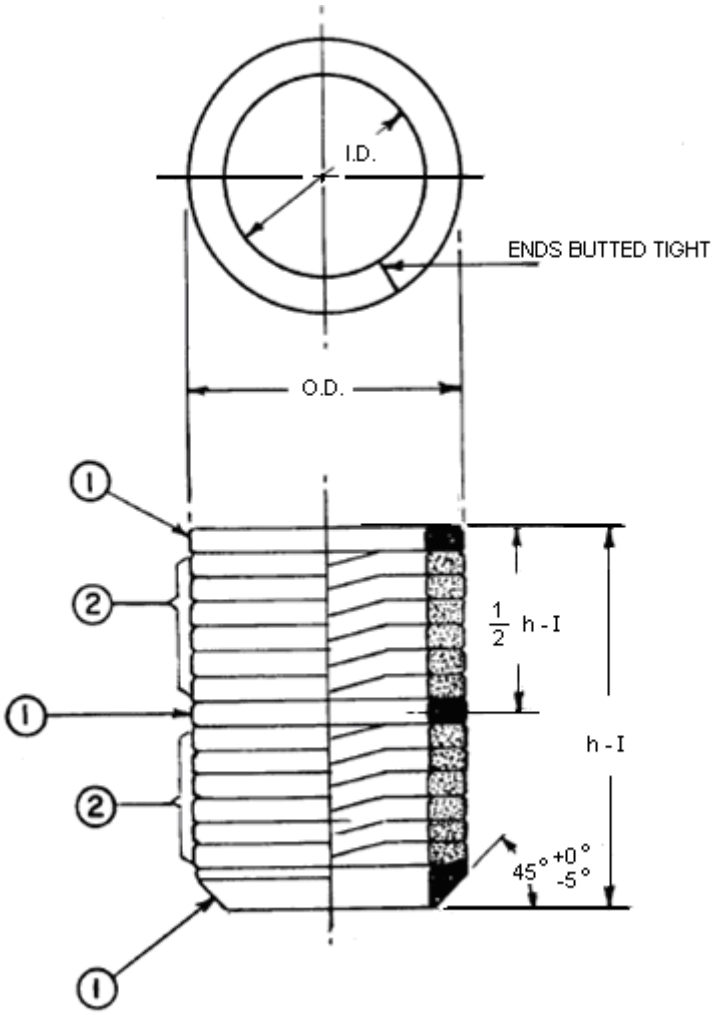


Figure 2 - Type I (sizes K to Z)

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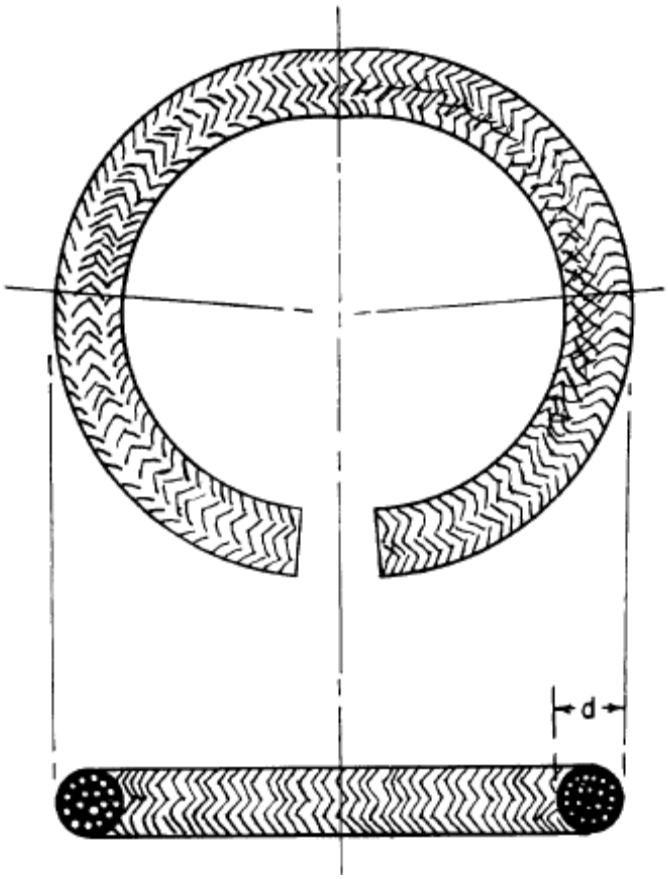
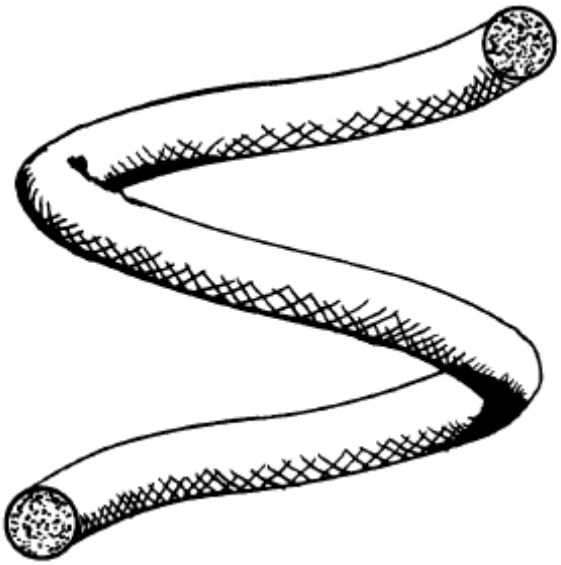


Figure 3 – Piece 1

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Figure 4 - Piece 2.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The packing covered by tnis specification is intended for use for stuffing tubes for electric cables (see table III) .

Table II - Application of packing types, class 2.

Stuffing tubes		Plugs and outlets	
NAVSHIPS Drawing or Specification Number	Packing type required	NAVSHIPS Drawing or Specification Number	Packing type required
MIL-S-24235/9 MIL-S-24235/10 MIL-S-24235/12	I	9-S-4799-L 9000-S6202-74438 MIL-S-24235/11	II size B
9-S-4436-L, ALT 9 or later	I	MIL-S-24235/13	II
9-S-5166-L ALT 10 or later	I	MIL-S-24235/14	II
9-S-5166-L ALT 9 or earlier	IA		

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, and size required (see 1.2, 3.1.2 and 3.2.3).
c. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Retainer packing
Sealing packing

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Navy - SH

Preparing activity:
DLA – IS

(Project 5330-2009-001)

Review activities:
Navy - AS, EC, OS

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